SIMATS ENGINEERING

ASSESSMENT TOOL 2: INDUSTRY PROBLEM SOLVING TASK

COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA1507

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1. RETAIL DATA OPTIMIZATION PROBLEM

Question:
A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behaviour data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

A big data architecture can be designed using the following layers:

1. Data Sources:
   Sales data, inventory data, online transactions, customer clicks, reviews and purchase history are collected from stores and the online platform.
2. Data Ingestion:
   Apache Kafka can be used to collect real-time data from different sources. Batch data can be uploaded using cloud storage services.
3. Data Storage:
   A data lake such as Amazon S3, Google Cloud Storage or Azure Data Lake can store large volumes of structured and unstructured data.
4. Data Processing:
   Apache Spark can process both real-time and historical data. Spark Streaming can be used for real-time inventory updates.
5. Analytics:
   Machine learning models such as Random Forest, XGBoost or time-series forecasting can be used for demand prediction. Collaborative filtering can be used for personalized product recommendations.
6. Visualization:
   Power BI or Tableau can display inventory levels, sales trends and demand forecasts.

Architecture Flow:

Data Sources → Kafka → Data Lake → Spark Processing → Machine Learning → Dashboard / Recommendations

Security:
Data should be protected using encryption, authentication, role-based access control and secure APIs.

Conclusion:
This architecture provides real-time inventory tracking, accurate demand forecasting and personalized marketing while remaining scalable for large retail data.

2. SOCIAL MEDIA FRAUD DETECTION PROBLEM

Question:
A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

A scalable big data solution can process social media activities in real time.

1. Data Collection:
   Posts, comments, likes, shares, user profiles and login activities are collected continuously.
2. Stream Processing:
   Apache Kafka can collect incoming events. Apache Spark Streaming or Apache Flink can process the data in real time.
3. Data Storage:
   A distributed data lake can store historical user activities. NoSQL databases such as MongoDB or Cassandra can store high-volume user data.
4. Machine Learning:
   Machine learning models can identify suspicious behavior. Random Forest, Logistic Regression and neural networks can classify fake accounts. NLP models can analyze text and identify possible misinformation.
5. Detection:
   Features such as unusual posting frequency, repeated content, suspicious login patterns and abnormal sharing behavior can be used to calculate a fraud score.

Architecture Flow:

Social Media Data → Kafka → Spark/Flink → ML Models → Fraud/Misinformation Score → Alert System

Security:
Access control, encryption, identity verification and continuous monitoring should be implemented.

Conclusion:
Stream processing provides fast detection, while machine learning improves the accuracy of identifying fake accounts and misinformation.

3. GOVERNMENT DATA PLATFORM PROBLEM

Question:
A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

A centralized and secure data governance strategy should be implemented.

1. Data Integration:
   Census, economic and geospatial data should be collected from different government departments and integrated into a common data platform.
2. Data Storage:
   A secure government data lake can store large datasets. Structured data can be maintained in relational databases.
3. Data Standards:
   Common formats, metadata standards, APIs and unique identifiers should be used to ensure interoperability between departments.
4. Data Governance:
   A data governance team should define data ownership, quality standards, access policies and retention rules.
5. Security:
   Sensitive information should be protected using encryption, authentication, authorization and role-based access control.
6. Privacy:
   Only authorized users should access sensitive information. Personal data should be anonymized or masked wherever required.
7. Data Quality:
   Data validation, duplicate removal and consistency checks should be performed regularly.

Architecture Flow:

Government Departments → Secure APIs → Data Integration → Government Data Lake → Analytics / Department Applications

Conclusion:
Strong governance, standardization, security and privacy controls will enable departments to share reliable data safely and efficiently.

4. BANKING FRAUD DETECTION PROBLEM

Question:
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

A real-time big data pipeline can be designed to detect fraudulent transactions quickly.

1. Data Sources:
   Transaction amount, account details, location, device information, transaction time and customer history are collected.
2. Data Ingestion:
   Apache Kafka can receive millions of transaction events in real time.
3. Stream Processing:
   Apache Flink or Spark Streaming can process transactions with low latency.
4. Data Storage:
   Historical transaction data can be stored in a data lake such as Amazon S3 or HDFS. A NoSQL database can be used for fast access to customer information.
5. Fraud Detection Algorithms:
   Random Forest, XGBoost and Logistic Regression can be used for classification. Anomaly detection can identify unusual transaction patterns.
6. Real-Time Decision:
   Each transaction receives a fraud probability score. High-risk transactions can be blocked or sent for manual verification.

Architecture Flow:

Transactions → Kafka → Flink/Spark → Feature Extraction → ML Model → Fraud Score → Approve / Block / Alert

Security:
Encryption, multi-factor authentication, access control and transaction monitoring should be implemented.

Conclusion:
The proposed pipeline provides low-latency fraud detection while using historical and real-time data to improve accuracy.

5. AGRICULTURE SMART FARMING PROBLEM

Question:
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

A precision agriculture system can use IoT, cloud computing and big data analytics.

1. Data Collection:
   IoT sensors collect soil moisture, temperature, humidity, nutrient and crop information. Drones provide images of crops and fields.
2. Data Ingestion:
   Sensor data can be transmitted using IoT gateways and collected through cloud-based services.
3. Data Storage:
   A cloud data lake can store sensor readings, drone images, weather data and historical crop information.
4. Data Processing:
   Apache Spark can process large amounts of agricultural data. Real-time processing can identify immediate changes in field conditions.
5. Analytics:
   Machine learning models can predict crop yield, detect crop diseases and recommend irrigation or fertilizer requirements.
6. Decision Support:
   Analytics dashboards can show soil conditions, crop health, weather trends and resource requirements to farmers.

Architecture Flow:

Sensors/Drones → IoT Gateway → Cloud Storage → Big Data Processing → ML Analytics → Farmer Dashboard → Decision

Benefits:
• Improved crop yield
• Reduced water usage
• Optimized fertilizer usage
• Early disease detection
• Better resource management

Security:
IoT devices should use authentication and encrypted communication. Access to agricultural data should be controlled.

Conclusion:
Big data analytics helps farmers make data-driven decisions, improve productivity and reduce the wastage of water, fertilizers and other resources.